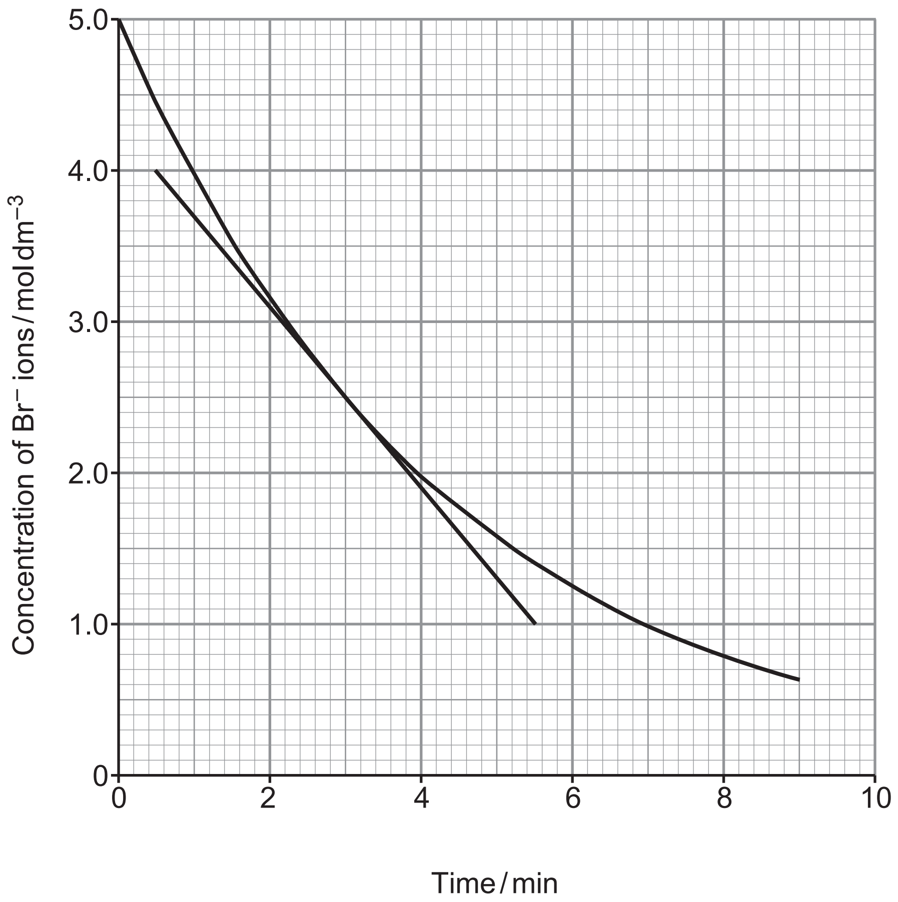
Examiner only

9. Bromide ions react with bromate(V) ions, BrO_3^-, as shown in the equation.

$$5Br^-(aq) \quad + \quad BrO_3^-(aq) \quad + \quad 6H^+(aq) \longrightarrow 3Br_2(aq) \quad + \quad 3H_2O(l)$$

Some students investigated the rate of this reaction by following the concentration of bromide ions over time. They plotted their results as shown.



Examiner only

(a) (i) The students calculated that the initial rate of the reaction was $1.20\ mol\ dm^{-3}\ min^{-1}$.

Use the tangent to the curve to calculate the rate of the reaction when the concentration of bromide ions had fallen to $2.50\ mol\ dm^{-3}$.

You **must** show your working. [2]

Rate = ... $mol\ dm^{-3}\ min^{-1}$

(ii) Use your answer to part (i) to suggest the relationship between the rate of this reaction and the concentration of bromide ions. [1]

....

....

(b) Suggest how the students could follow the rate of this reaction. Explain your answer. [2]

....

....

....

(c) In the experiment, the concentration of bromide ions fell from $5.0\ mol\ dm^{-3}$ to $2.0\ mol\ dm^{-3}$ over the first 4 minutes. The initial concentration of bromate(V) ions was $1.0\ mol\ dm^{-3}$.

$$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \longrightarrow 3Br_2(aq) + 3H_2O(l)$$

Using the equation, deduce the concentration of bromate(V) ions after 4 minutes. [1]

Concentration of bromate(V) ions = ... $mol\ dm^{-3}$

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